

SAQIB ALI

ELECTRICAL ENGINEERING STUDENT

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PROFILE

A driven engineering student with a solid grounding in Real-time Control Systems and Signal Processing. Experienced in creating adaptive control solutions for high-inertia systems. Proficient in Fusion 360 for 3D modeling and design, and skilled in conducting Finite Element Analysis (FEA) to evaluate product performance and durability. Enthusiastic about emerging technologies, such as Generative AI and aerospace applications, and eager to apply technical skills in a challenging engineering position.

EDUCATION

Zakir Husain College of Engineering & Technology (ZHCET), Aligarh B.Tech in Electrical Engineering
Sep 2023 - Jun 2027

Syedna Tahir Saifuddin (STS) School, AMU
Senior Secondary School (PCM)

PROFESSIONAL EXPERIENCE

Electrical Design Intern

Orbitalink Private Limited(in collaboration with AMU Innovation Foundation) | Aligarh, UP

- Developed and implemented an Adaptive PID Controller designed to achieve stabilization and precise control of a high-inertia antenna tracking system.
- Applied advanced control theory principles, system modeling, and performance optimization for critical satellite communication infrastructure.

Ground station TeamMember

SS AMU SAT | Aligarh, UP

- Played a key role in developing and building the 3D model chassis for the antenna tracker.
 - Conducted tests on antenna tracking mechanisms.
 - Assisted with ground station operations and satellite tracking protocols.
 - Helped maintain and calibrate communication equipment to guarantee reliable data transmission.
- Acquired hands-on experience in signal reception, hardware troubleshooting, and system diagnostics.

SKILLS

Hard Skills & Expertise

- Real-time Control Systems, Signal Processing, and Analysis
- Adaptive PID Control, System Modeling, and Optimization
- Development Skills: Proficient in Python, Full Stack Web Development, and Data Science (including NumPy, Pandas, Matplotlib, and Seaborn), as well as sensor interfacing and I2C protocols, with expertise in C++ for hardware control.

- Engineering Expertise: Skilled in Aircraft Design Principles and Mechanical Product Design utilizing FUSION 360. Control Block Design, Simulink

Soft Skills

- Communication
- Problem Solving
- Adaptability
- Team Collaboration
- Detail-Oriented

Projects

Weather Dashboard Application

[Live Deployment](#) \ [Source Code \(GitHub\)](#) OpenWeatherMap API, Node.js, Axios

- Deployed on Render using OpenWeatherMap API, Node.js, and Axios.
- Developed a responsive weather forecasting app with Modern JavaScript and CSS3.
- Integrated OpenWeatherMap API for real-time data on temperature, humidity, and wind speed globally.
- Created a robust search function for querying weather conditions worldwide.
- Established error handling protocols to manage API failures and invalid inputs.

Simulink PID Control System Modeling

File: [PID_SIMULATIONV1.slx](#) | Platform: Simulink

- Designed and simulated a dynamic control loop in Simulink to model the behavior of a physical plant.
- Verified controller performance against various setpoint changes and noise inputs.

Simulink Modeling and Analysis of a DC-DC Boost Converter

File: [Model_Boost.slx](#) | Platform: Simulink

- Designed and simulated a DC-DC Boost Converter circuit in Simulink as part of my college laboratory work, showcasing the ability to step up a DC input voltage.
- Analyzed the converter's steady-state and transient responses under varying duty cycles and load conditions.
- This project demonstrates proficiency in Power Electronics, Circuit Simulation, and Switch-Mode Power Supply (SMPS) fundamentals.

3D Chassis Design for Motorized Tracking Unit

- Crafted the 3D model of the chassis for the motorized tracking unit using Fusion 360, optimizing the design for rigidity and minimal weight while ensuring proper placement for the motor, gearbox, and sensors.
 - Implemented the core PID control logic for the two-axis antenna positioning system, which is crucial for maintaining a lock on a satellite signal.
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CERTIFICATIONS

- Certificate of Participation in MATLAB Simulation and Control (IEEE Student Branch, AMU)
- Advanced Aircraft Design Workshop (Bharat Space Education Research Centre)
- Certificate of Participation (SS AMU SAT)

ACTIVITIES

SSAMUSAT Antenna Tracker Development Team

- Position Held: Member of the Ground Station Team
- Tenure: September 2024 – Present
- Role Description: As an enthusiastic participant in the university's satellite development team, I am committed to the meticulous design and construction of a sophisticated ground station antenna tracker, ensuring high precision and reliability in satellite communication.
- Advanced Aircraft Design Workshop (2024): Participated in a comprehensive workshop focused on the intricacies of aircraft design, enhancing both theoretical understanding and practical skills.
- SS AMU SAT Certificate of Participation (2024): Received certification for active involvement in the workshop conducted by SS AMU SAT project.
- Google AI Training: Completed certifications in Develop GenAI Apps with Gemini and Streamlit. and Prompt Design in Vertex AI (Issued May 2024).
- IEEE Student Branch, AMU:
 - Attained a certificate for participation in Matlab Simulation and Control (Issued Oct 2025).
 - Completed Introduction to Cybersecurity, acquiring fundamental knowledge in Cybersecurity
- Visit to Indian Mobile Congress (31 October 2024)
Represented Orbitalink and showcased the Field Unit at the event.

